

Grade 3 Ontario Curriculum Algebra

Bug Busters

Unit Overview: Bug Busters

Grade: Grade 3 | **Strand:** Algebra

Target Expectations: C3.1, C3.2

Learning Goal: Children will write and execute code with sequential, concurrent, and repeating events; read and alter buggy code; and describe how changes affect outcomes.

Lesson	Lesson Title	Learning Goal
Day 1	Concurrent Events	I can identify and write code with concurrent events.
Day 2	Writing Concurrent Code	I can write code with concurrent events.
Day 3	Finding Bugs	I can find a bug in existing code and explain it.
Day 4	Fixing Bugs	I can fix a bug by altering the code.
Day 5	Bug Buster Capstone	I can write code, plant a bug, and check my partner's fix.

Lesson 1: Concurrent Events

Goal: Some events happen AT THE SAME TIME (concurrent) instead of one after another.

Expectation: C3.1

Learning Goal: "I can identify and write code with concurrent events."

Materials: M1_grid_mat, M2_code_cards, M4_robot_markers, M5_concurrent_chart

1. Minds On (12 mins): Buzz Introduces Concurrent

Buzz: Two robots can move AT THE SAME TIME. That's concurrent! Buzz opens: 'Read by ROW, not by LANE!'

Bring children to the carpet. Buzz puppet. 'Friends — I'm Buzz the Bug-Buster Beetle. Today we learn CONCURRENT events.'

2. Action (25 mins): Concurrent Stations

- Whole group: identify 4 examples as concurrent or sequential.
- Three stations.
- Stretch challenge at Station C: write code where Robot 1 has 5 steps and Robot 2 has 3 steps.
- Discuss the time-step idea.
- Circulate to record formative evidence on the concurrent tracker.
- STRATEGY COMPARE: Two pairs share their reads of the same 2-lane code.

3. Consolidation (13 mins): Buzz's Concurrent Brag

Discuss:

- What does CONCURRENT mean?
- How is it different from SEQUENTIAL?
- Real-life concurrent example?
- What does each row of 2-lane code mean?
- Buzz's preview: tomorrow we WRITE concurrent code (Lesson 2 — C3.1).

Exit: Buzz's refrain — class chants: 'BUZZ says: Read by ROW, not by LANE!' (3x).

Lesson 2: Writing Concurrent Code

Goal: Concurrent code uses parallel lanes. Two actions in the same time-step run together.

Expectation: C3.1

Learning Goal: "I can write code with concurrent events."

Materials: M1_grid_mat, M2_code_cards, M4_robot_markers, M5_concurrent_chart

1. Minds On (12 mins): Buzz Writes 2-Lane Code

Buzz writes code: Robot 1 forward + Robot 2 forward at the SAME time. Buzz adds a loop. Buzz opens: 'every LANE, every STEP, gets an ACTION!'

Bring children to the carpet. Buzz + Patch puppets. 'Yesterday we IDENTIFIED concurrent. Today we WRITE 2-lane code with a loop.'

2. Action (25 mins): Concurrent Writing Stations

- Whole group: write 2 codes together.
- Three stations.
- Stretch challenge at Station C: write code where R1 has 6 steps and R2 has 3 steps.
- Discuss the time-step model.
- Circulate to record formative evidence on the writing tracker.
- STRATEGY COMPARE: Two pairs share their concurrent code that produces the same outcome.

3. Consolidation (13 mins): Buzz's Writing Brag

Discuss:

- What does CONCURRENT mean in code?
- How does a loop work in concurrent code?
- What if R1 and R2 finish at different times?
- Show me your code's ending positions.

Exit: Buzz's writing refrain — 'every LANE, every STEP, gets an ACTION!' (3x).

Lesson 3: Finding Bugs

Goal: A BUG is a problem in code that makes the outcome wrong.

Expectation: C3.2

Learning Goal: "I can find a bug in existing code and explain it."

Materials: M1_grid_mat, M2_code_cards, M3_bug_cards, M4_robot_markers

1. Minds On (12 mins): Buzz Hunts a Bug

Buzz: This code says R1 should end at (3,0). But it ends at (4,0). Where's the BUG? Buzz opens: 'TRACE it, FACE it, REPLACE it!' Yesterday we WROTE code — today we HUNT bugs.

Bring children to the carpet. Buzz + Patch puppets. 'Friends — today we BUST BUGS. A bug is a problem in code that makes the outcome WRONG.'

2. Action (25 mins): Bug-Finding Stations

- Whole group: find bugs in 4 codes.
- Three stations.
- Stretch challenge at Station C: find a bug in a 2-lane code where one lane is buggy.
- Discuss bug-finding strategy.
- Circulate to record formative evidence on the finding tracker.
- STRATEGY COMPARE: Two children share their bug-finding approaches — Strategy A 'trace from the start' vs Strategy B 'jump to the suspicious cell'.

3. Consolidation (13 mins): Buzz's Finding Brag

Discuss:

- What's a BUG?
- Three types of bugs?
- What's the bug-finding strategy?
- Why do bugs hide?

Exit: Buzz's bug-busting chant — 'TRACE it, FACE it, REPLACE it!' (3x).

Lesson 4: Fixing Bugs

Goal: We fix bugs by altering the code. Change the wrong step, the wrong number, or the wrong order.

Expectation: C3.2

Learning Goal: "I can fix a bug by altering the code."

Materials: M1_grid_mat, M2_code_cards, M3_bug_cards, M4_robot_markers

1. Minds On (12 mins): Buzz and Patch Fix Bugs

Patch: I found the bug. Buzz: Now we FIX it. Alter the code. Buzz opens: 'FIND it, ALTER it, TEST it!'

Bring children to the carpet. Buzz + Patch puppets. 'Yesterday we FOUND bugs. Today we FIX them.'

2. Action (25 mins): Bug-Fixing Stations

- Whole group: fix bugs in 4 codes.
- Three stations.
- Stretch challenge at Station C: fix a bug in 2-lane code where the bug is in one lane.
- Discuss the fix-and-verify pattern.
- Circulate to record formative evidence on the fixing tracker.
- STRATEGY COMPARE: After Step 2, each pair shares HOW they fixed their bug — line-edit vs row-rewrite vs full-rewrite.

3. Consolidation (13 mins): Buzz's Fixing Brag

Discuss:

- What does ALTER mean?
- What's the find-fix-verify pattern?
- Why do we verify after fixing?
- What if our fix introduces a NEW bug?

Exit: Buzz's fixing chant — 'FIND it, ALTER it, TEST it — that's DEBUGGING!' (3x).

Lesson 5: Bug Buster Capstone

Goal: We can write our own code AND plant a bug for a partner to find.

Expectation: C3.1, C3.2

Learning Goal: "I can write code, plant a bug, and check my partner's fix."

Materials: M1_grid_mat, M2_code_cards, M4_robot_markers, M6_capstone_template

1. Minds On (12 mins): Buzz and Patch's Final Showcase

Now you write code AND plant a bug. Your partner finds and fixes it. Buzz + Patch open: 'TRACE it, FACE it, REPLACE it! FIND it, ALTER it, TEST it!'

Bring children to the carpet. Buzz + Patch together. 'Friends — today you become BUG BUSTERS. Write your own 2-lane code with concurrent events. Plant a bug. Swap with a partner. They debug. Verify together.'

2. Action (25 mins): Bug Buster Capstone

- Hand out capstone templates.
- Children plant ONE bug in their code.
- Stretch challenge: plant TWO bugs.
- Swap with a partner.
- Whole-class share: 5-6 children share aloud.

3. Consolidation (13 mins): Buzz's Final Brag and Certificate Ceremony

Discuss:

- What's the most surprising bug you saw today?
- How did you plant your bug?
- How did you fix your partner's bug?
- What's something in coding you want to learn next?

Exit: Capstone refrain — Buzz + Patch call: 'TRACE it, FACE it, REPLACE it! FIND it, ALTER it, TEST it!' Class chants 3x.

Name: _____

Date: _____

Buzz's Concurrent Cases

Learning Goal: I can identify and write code with concurrent events.

Instructions for the Student:

1. Sort 8 scenarios as CONCURRENT or SEQUENTIAL.
2. Trace 2 existing 2-lane codes.
3. Write your own 2-lane code with 4 time-steps.

Part 2: Trace 2-Lane Code

Trace 2 existing 2-lane codes. Predict where each robot ends.

[illegible]

Name: _____

Date: _____

Buzz's Concurrent Writing

Learning Goal: I can write code with concurrent events.

Instructions for the Student:

1. Write a 4-step 2-lane code where both robots are synchronised.
2. Write a 2-lane code where one lane has a loop.
3. Write your own 2-lane code (4-6 time-steps).

Part 1: Synchronised Code

Write a 4-step 2-lane code where both robots are synchronised.

[illegible]

Date: _____

Learning Goal: I can find a bug in existing code and explain it.

1. Find the bug in 6 buggy codes.
2. Trace 2 codes step by step.
3. Write your own buggy code.

Trace 2 codes step by step. Where does it diverge from expected?

[illegible]

Date: _____

Buzz's Fix It

Learning Goal: I can fix a bug by altering the code.

Instructions for the Student:

1. Fix 6 buggy codes.
2. Identify and fix bugs in 4 short codes (mixed types).
3. Take a partner's bug.

Part 1: Fix Six Codes

Fix 6 buggy codes. Show your alteration. Verify by re-tracing.

[illegible]

Name: _____

Date: _____

My Bug Buster Project

Learning Goal: I can write code, plant a bug, and check my partner's fix.

Instructions for the Student:

1. Write a 2-lane concurrent code (4-6 time-steps).
2. Plant ONE bug.
3. Swap with a partner.

Part 1: My Code

Write a 2-lane concurrent code (4-6 time-steps).

2-Lane Coding Grid Mat

(M1_grid_mat)

Letter-size mat with 2 robot lanes for concurrent code.

Robot Grid Mat

	0	1	2	3	4	5	6	7
0								
1								
2								
3								
4								
5								

PRINT SPECS

Size: letter landscape

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 per pair

Lessons: 1, 2, 3, 4, 5

Prep: ~12 min

TEACHER PREP

1. Print 1 mat per pair.

2. Laminate.

3. Hand out Lesson 1.

4. Reuse Lessons 2-5.

5. Use dry-erase.

Code Step Cards

(M2_code_cards)

30 step cards (forward, turn, repeat).

Code Cards

MOVE 1 RIGHT	MOVE 1 LEFT	MOVE 1 UP	MOVE 1 DOWN
TURN LEFT	TURN RIGHT	REPEAT 3	STOP

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 5

Laminate: yes

USE IN CLASS

Quantity: 30 cards

Lessons: 1, 2, 3, 4, 5

Prep: ~25 min

TEACHER PREP

1. Print 5 pages on cardstock.

2. Laminate.

3. Cut into 30 cards.

4. Sort by type.

5. Distribute Lessons 1-5.

Buggy Code Cards

(M3_bug_cards)

12 cards with planted bugs.

Bug Cards

wrong direction	skipped step	wrong number	missing repeat
extra step	swapped pair	wrong start	wrong end

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

USE IN CLASS

Quantity: 12 cards

Lessons: 3, 4

Prep: ~12 min

TEACHER PREP

1. Print 2 pages on cardstock.

2. Laminate.

3. Cut into 12 cards.

4. Sort by bug type.

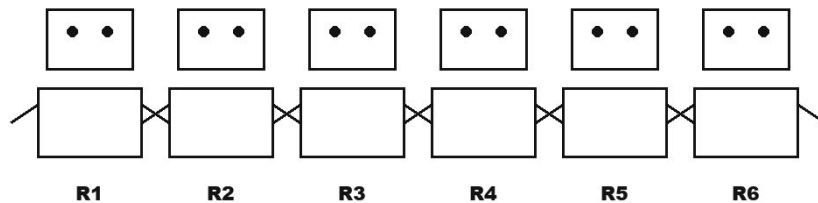
5. Distribute Lessons 3-4.

Robot Markers (2 colours)

(M4_robot_markers)

10 robot tokens, 2 colours for 2-lane code.

Robot Tokens



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 10 tokens

Lessons: 1, 2, 3, 4, 5

Prep: ~8 min

TEACHER PREP

1. Print 1 page on cardstock.
2. Cut into 10 tokens.
3. Sort into pair sets.
4. Have ready Lessons 1-5.
5. Optional: laminate.

Concurrent vs Sequential Chart

(M5_concurrent_chart)

Anchor chart contrasting concurrent and sequential.

Equation Table

Step	Robot A	Robot B

PRINT SPECS

Size: letter portrait
Colour: bw
Pages per set: 4
Laminate: yes

USE IN CLASS

Quantity: 1 chart
Lessons: 1, 2
Prep: ~15 min

TEACHER PREP

- 1. Print 4 pages and tile.
- 2. Laminate.
- 3. Tape to wall.
- 4. Reference Lessons 1-2.
- 5. Roll for re-use.

Bug Buster Capstone Template

(M6_capstone_template)

Letter-size template for Lesson 5 capstone.

Bug Buster Capstone Template	
1	2-Lane Code
2	Plant Bug
3	Partner Fix
4	Reflection

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: no

USE IN CLASS

Quantity: 1 per child

Lessons: 5

Prep: ~12 min

TEACHER PREP

1. Print 1 per child.

2. Hand out Lesson 5.

3. Children fill in.

4. Collect for assessment.

5. Display.

Loop Cards

(M7_loop_cards)

8 cards: Repeat 2, 3, 4, 5.

Loop Cards

REPEAT 3 [STEP]	REPEAT 5 [STEP]
REPEAT 2 [TURN, STEP]	REPEAT 4 [STEP, STEP]
WHILE NOT END	REPEAT UNTIL FOUND

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 8 cards

Lessons: 2, 3, 4, 5

Prep: ~10 min

TEACHER PREP

1. Print 1 page on cardstock.

2. Laminate.

3. Cut into 8 cards.

4. Distribute Lessons 2-5.

5. Optional: extras.

Code Alteration Cards

(M8_alter_cards)

8 cards each with original code and alteration instruction.

Alter Cards

swap two steps	add a STOP
remove a step	add a TURN
double the loop	halve the loop

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 8 cards

Lessons: 4

Prep: ~10 min

TEACHER PREP

1. Print 1 page on cardstock.

2. Laminate.

3. Cut into 8 cards.

4. Distribute Lesson 4.

5. Optional: extras.

Bug Type Anchor Chart

(M9_bug_chart)

Chart of common bug types: wrong step, wrong number, wrong order.

Equation Table

Code	Bug	Fix

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 4

Laminate: yes

TEACHER PREP

1. Print 4 pages and tile.

2. Laminate.

3. Tape to wall.

4. Reference Lessons 3-4.

5. Roll for re-use.

USE IN CLASS

Quantity: 1 chart

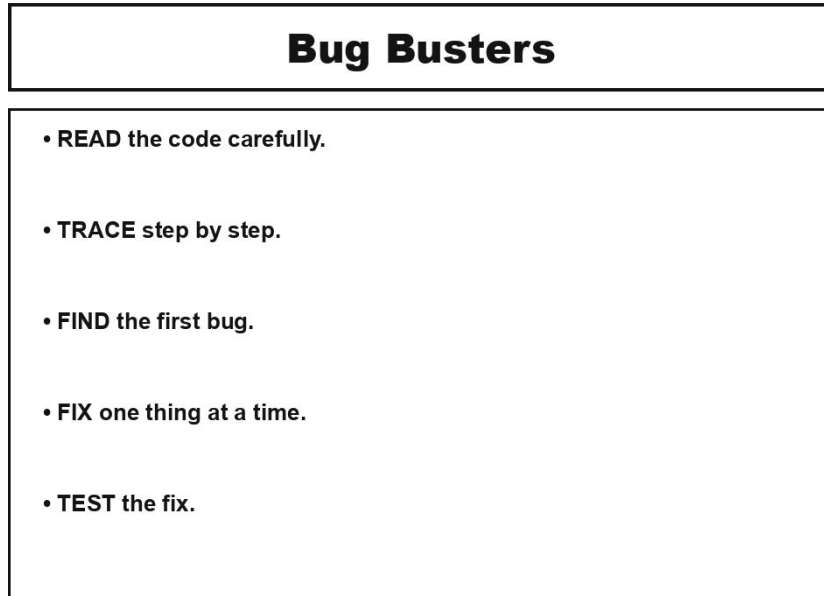
Lessons: 3, 4

Prep: ~15 min

Bug Busters Anchor Poster

(M10_buster_poster)

Anchor poster of Buzz and Patch.



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 4

Laminate: yes

TEACHER PREP

1. Print 4 pages and tile.

2. Laminate.

3. Tape to wall.

4. Reference all week.

5. Roll for re-use.

USE IN CLASS

Quantity: 1 poster

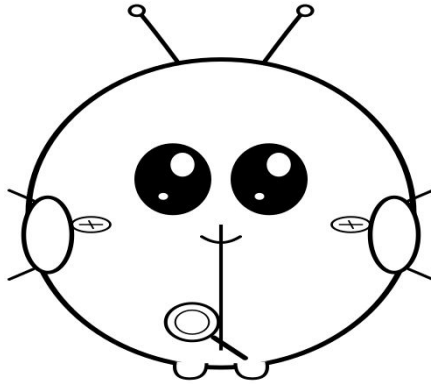
Lessons: 1, 2, 3, 4, 5

Prep: ~18 min

Buzz the Bug-Buster Beetle

(char_buzz_puppet)

Printable Buzz illustration on a popsicle stick.



Buzz the Bug-Buster Beetle

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 puppet

Lessons: 1, 2, 3, 4, 5

Prep: ~10 min

TEACHER PREP

1. Print on cardstock.

2. Cut.

3. Glue popsicle stick.

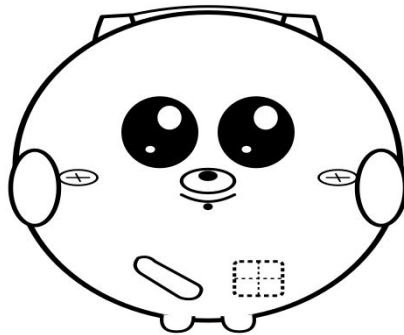
4. Optional: laminate.

5. Store flat.

Patch the Repair Pup

(char_patch_puppet)

Printable Patch illustration on a popsicle stick.



Patch the Repair Pup

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 puppet

Lessons: 2, 3, 4, 5

Prep: ~10 min

TEACHER PREP

1. Print on cardstock.

2. Cut.

3. Glue popsicle stick.

4. Optional: laminate.

5. Store flat.

Name: _____

Date: _____

Summative Assessment Rubric

Expectation	Level 1 (Beginning)	Level 2 (Developing)	Level 3 (Achieving)	Level 4 (Exceeding)
C3.1 solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events	Cannot write 2-lane code with concurrent events.	Writes 2-lane code with prompts and supports.	Writes complete 2-lane code with concurrent and repeating events.	Writes 2-lane code with multiple lanes; predicts all outcomes.
C3.2 read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes	Cannot find bugs or alter code.	Finds bugs with prompts and supports.	Traces, finds bugs, alters to fix, verifies.	Identifies bug type; predicts cascading fix effects.

Junior Bug Buster Certificate

This certificate is proudly presented to:

For successfully completing the 5-Day Bug Busters unit and demonstrating the skills of a Junior Bug Buster. Buzz and Patch congratulate the student.

By completing this unit, this student has demonstrated:

- Identifying concurrent vs sequential events
- Writing 2-lane concurrent code
- Finding bugs in existing code
- Altering code to fix bugs
- Inventing a capstone bug-busting project

Marketplace Listing

A 5-day Ontario Grade-3 unit on coding without screens (C3.1-C3.2) framed around 'Bug Busters' — Buzz the Bug-Buster Beetle and Patch the Repair Pup.

Price: \$9.99 CAD

Pages: ~32

Classroom time: 250 min

Teacher prep: ~90 min

Top tags: grade-3, math, ontario-curriculum, coding, algebra, concurrent, without-screens, canadian

What's included:

- 5 fully-scripted lesson plans with 50-minute durations and 250 minutes of total teaching time
- 5 student worksheets aligned to each lesson
- 1 mid-unit formative assessment (Bug Check-In)
- 10 printable teacher manipulatives plus 2 character puppets: 2-lane grid mat, code step cards (30), buggy code cards (12), robot markers, concurrent vs...
- 1 end-of-unit reflection sheet with 5 self-assessment prompts
- 4 class observation trackers: 1 diagnostic + 3 formative
- 1 two-row summative rubric covering C3.1-C3.2 across 4 levels
- 1 summative-administration guide tying the rubric to Lesson 5 capstone
- Junior Bug Buster certificate (one per child)
- Detailed expanded walkthroughs covering common misconceptions
- Differentiation tracks built into every lesson